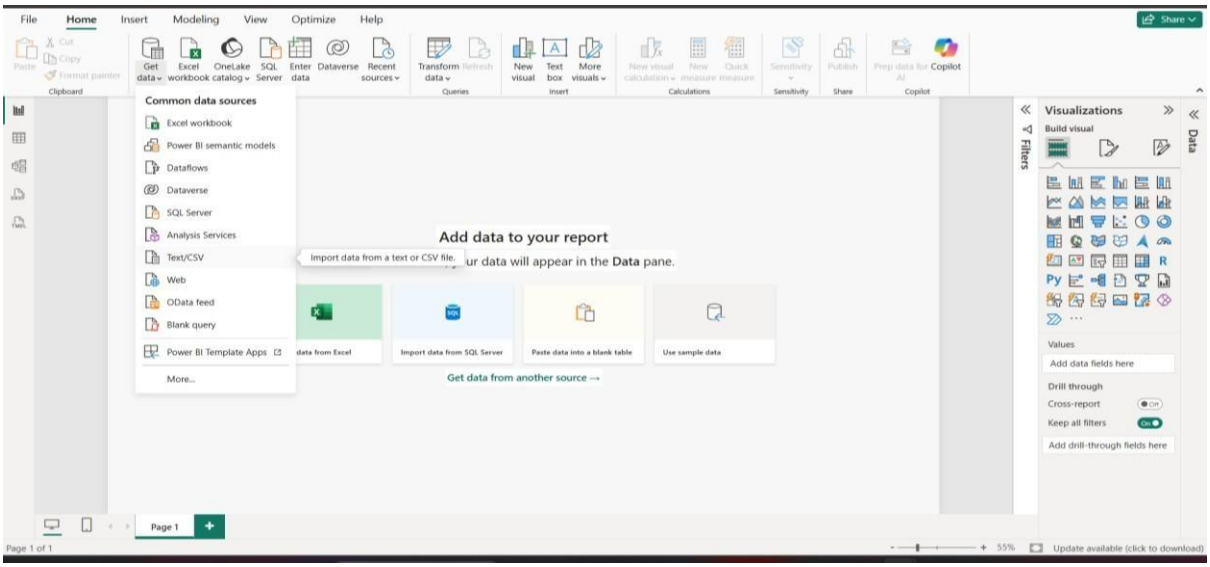
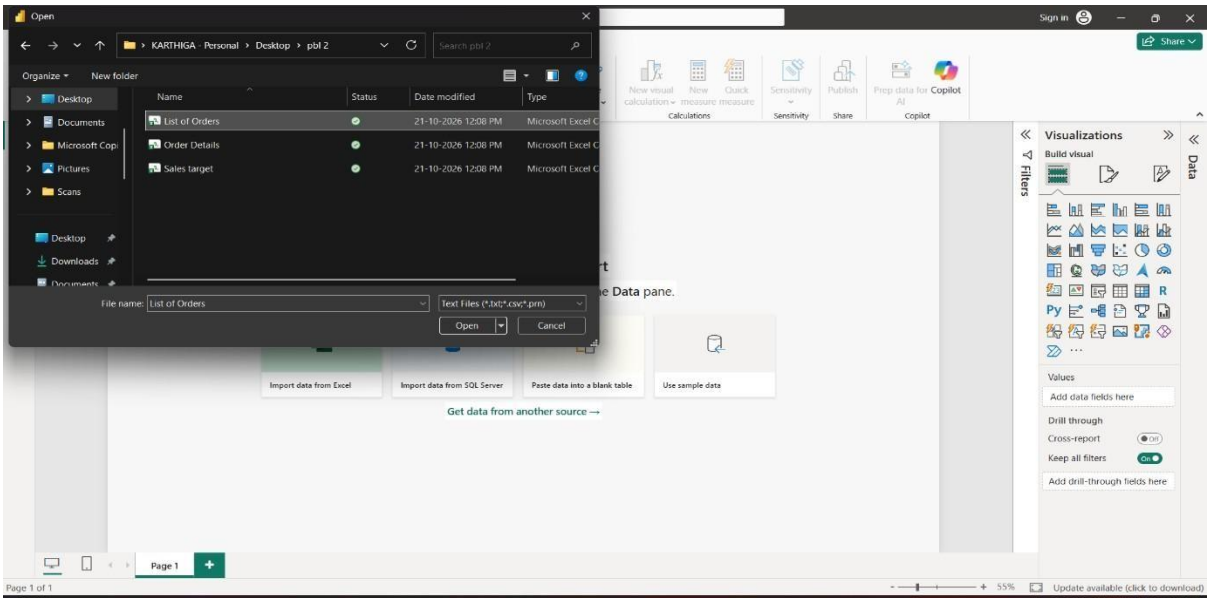
Power BI Assignment 2 – DAX & Data Visualization

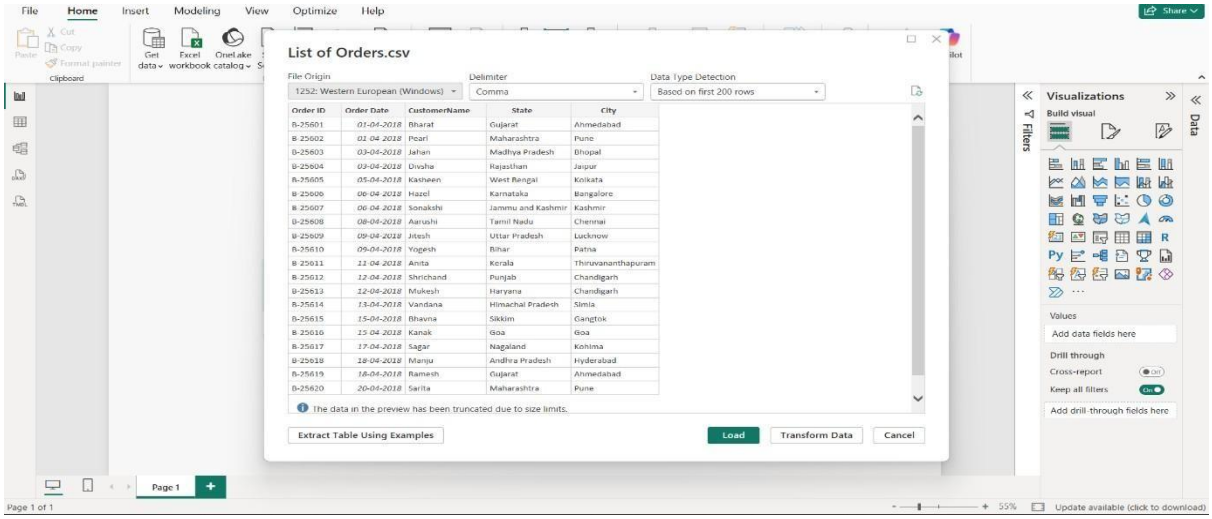
Here , the three E- Commerce Sales Analysis datasets are given CSV format .

- ★ **List of Orders.csv**
- ★ **Order Details.csv**
- ★ **Sales target.csv**

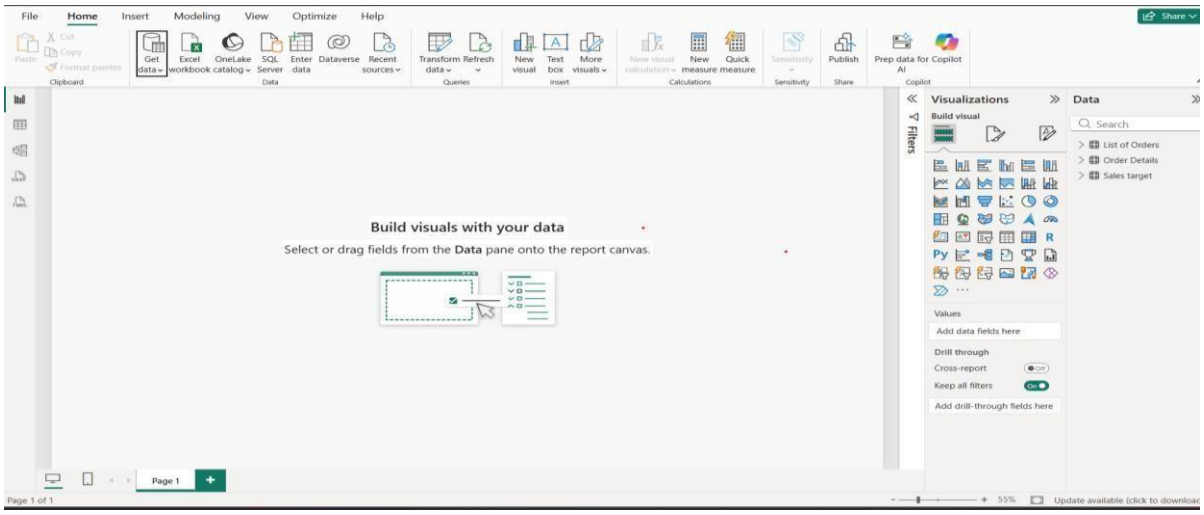
These files are imported into the Power BI and into the Power Query Editor.

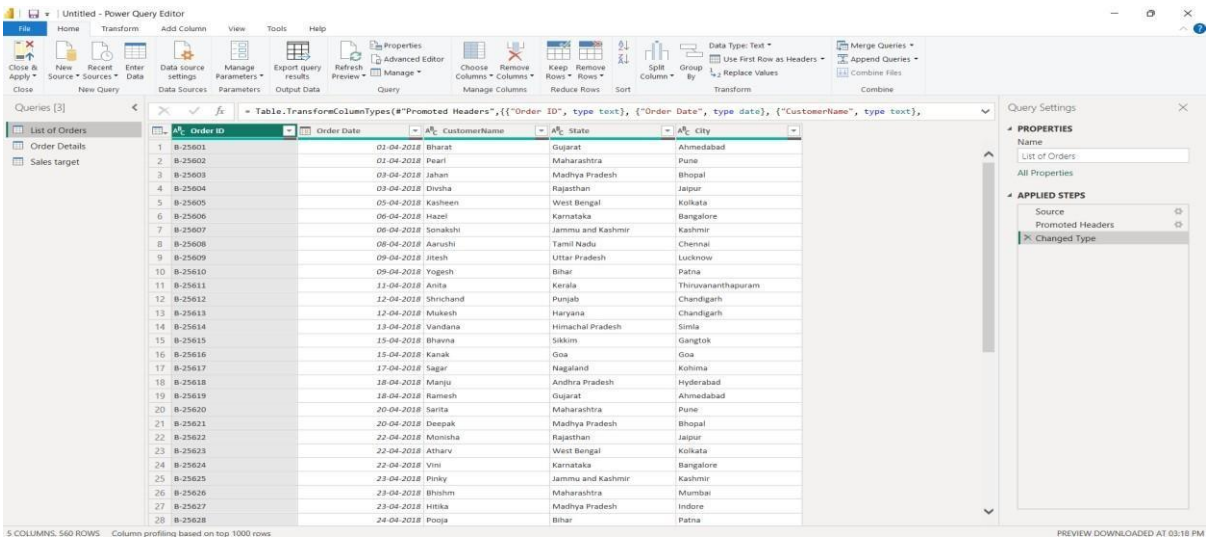




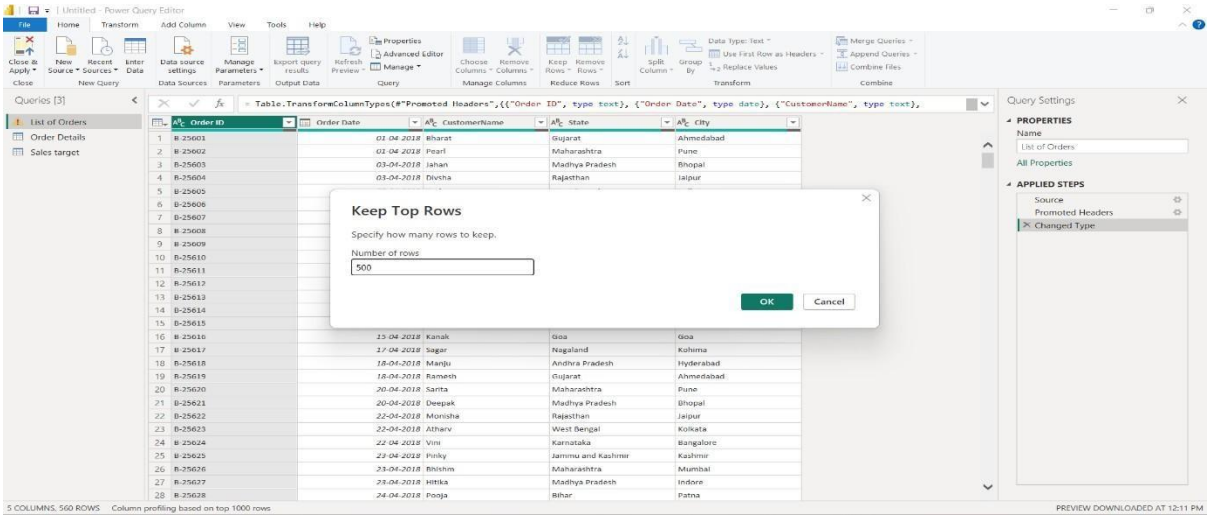


The same procedure is repeated for the remaining 2 CSV files and are imported into the PowerBI and Power Query Editor.

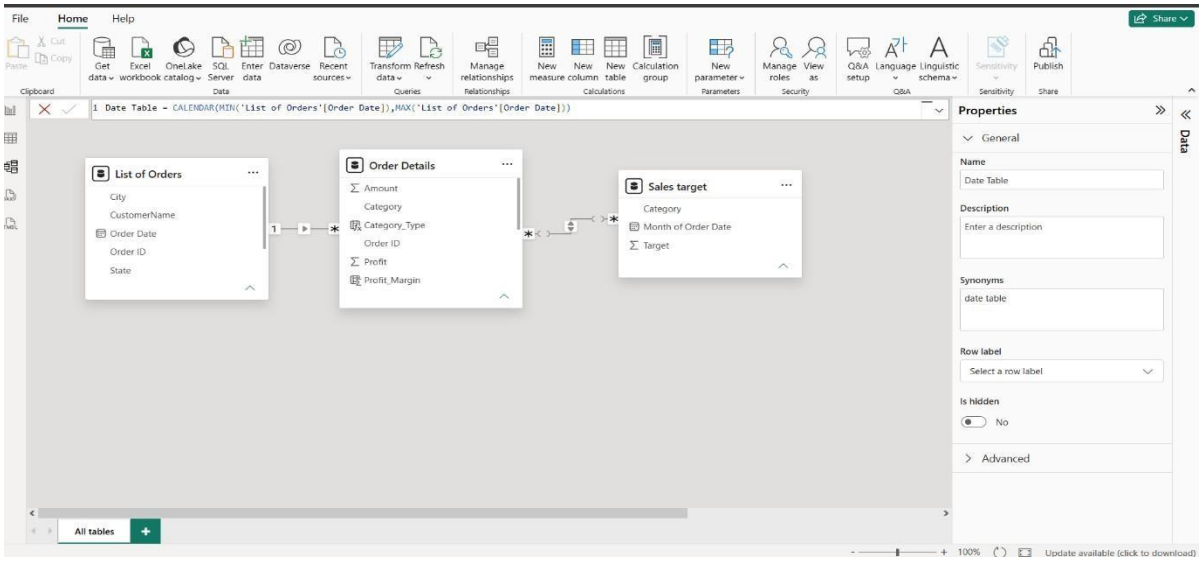




In the LIST OF ORDERS table , there are 560 rows totally with 60 NULL valued rows at the bottom . So, we can remove those NULL valued rows and keep top 500 rows for further Analysis.



DATA MODELING



The Relationship between the given 3 datasets are established.

List Of Orders (Order Id) ----- Order Details (Order Id)

Cardinality – Many to One Cross Filter Directions – Single

Order Details (Category) ----- Sales Target (Category)

Cardinality – Many to Many Cross Filter Directions – Both

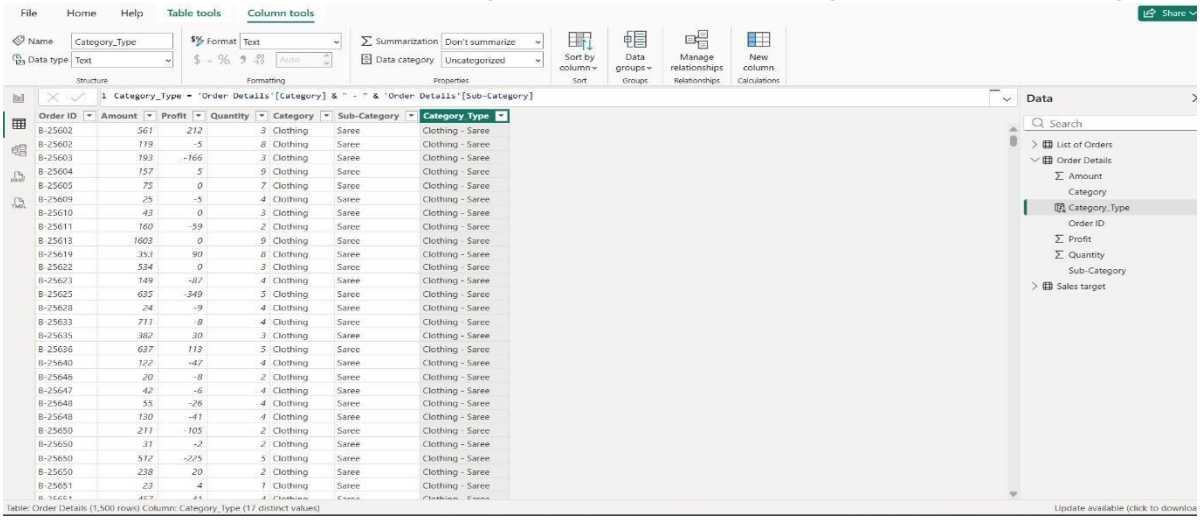
CALCULATED COLUMNS

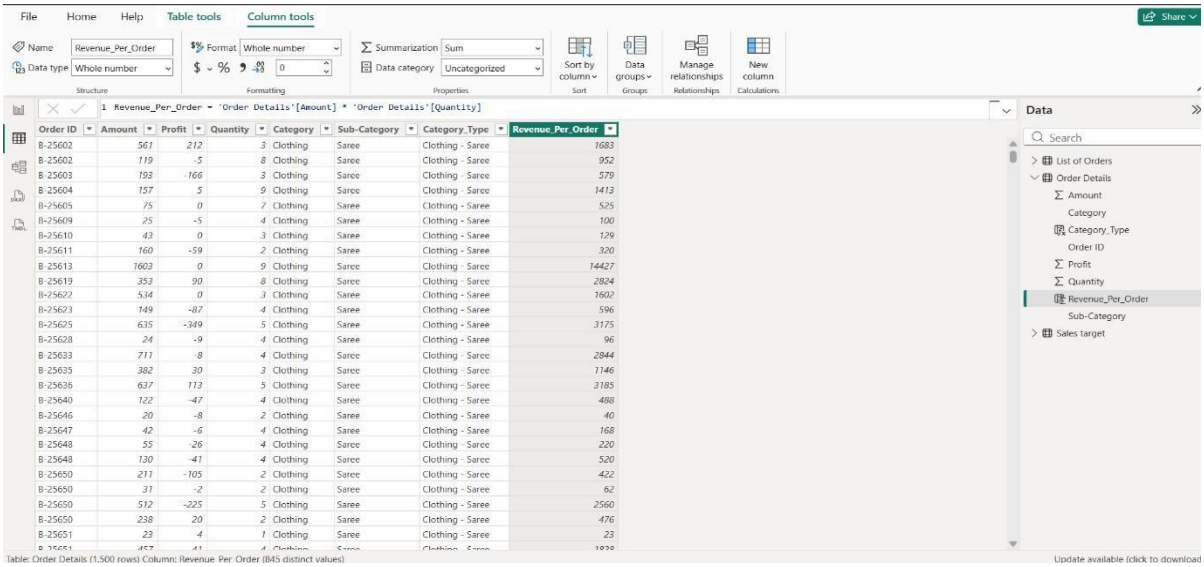
In the ORDER DETAILS table , the 3 columns are created and are named as CATEGORY_TYPE , REVENUE_PER_ORDER , SALES_CATEGORY .

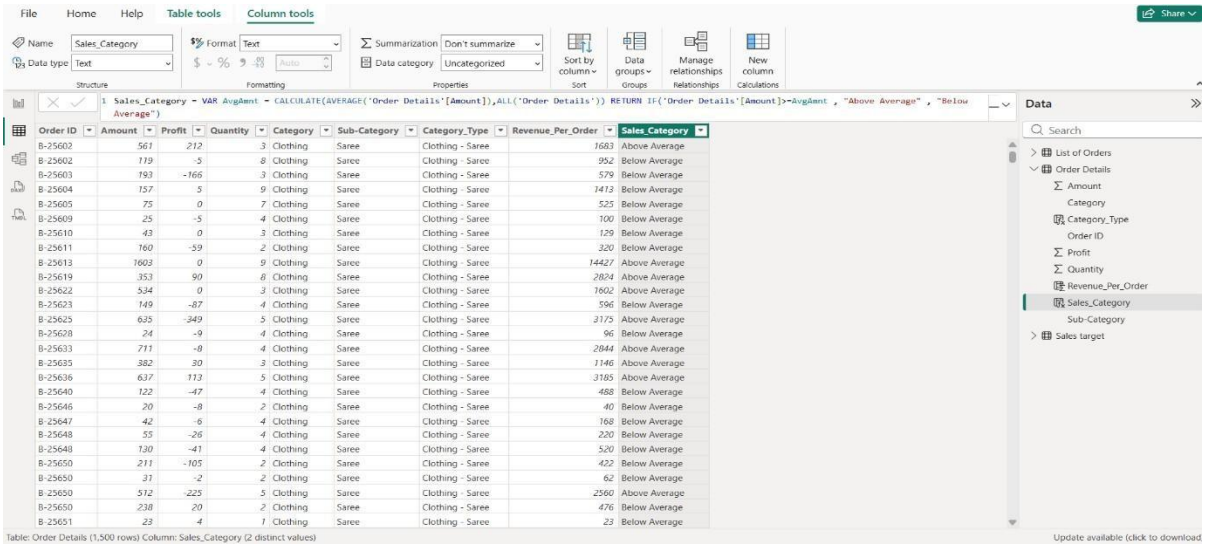
CATEGORY _TYPE = 'Order Details'[Category] & " - " & 'Order Details'[SubCategory]

REVENUE_PER_ORDER = 'Order Details'[Amount] * 'Order Details'[Quantity]

SALES_CATEGORY = VAR AvgAmnt = CALCULATE(AVERAGE('Order Details'[Amount]),ALL('Order Details')) RETURN IF('OrderDetails'[Amount]>=AvgAmnt , "Above Average" , "Below Average")







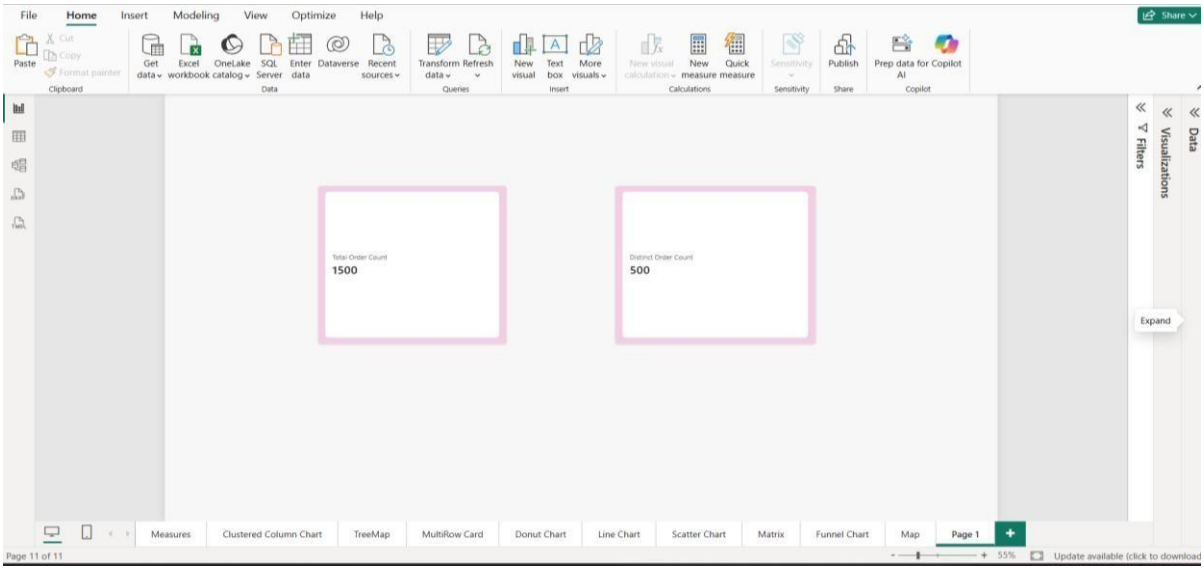
CALCULATED MEASURES

1. CALCULATE ORDER COUNT

In the ORDER DETAILS Table , the COUNTROWS () and DISTINCTCOUNT () functions are used to find the total order count and the distinct order count.

TOTAL ORDER COUNT = COUNTROWS('Order Details')

DISTINCT ORDER COUNT = DISTINCTCOUNT('Order Details'[Order ID])

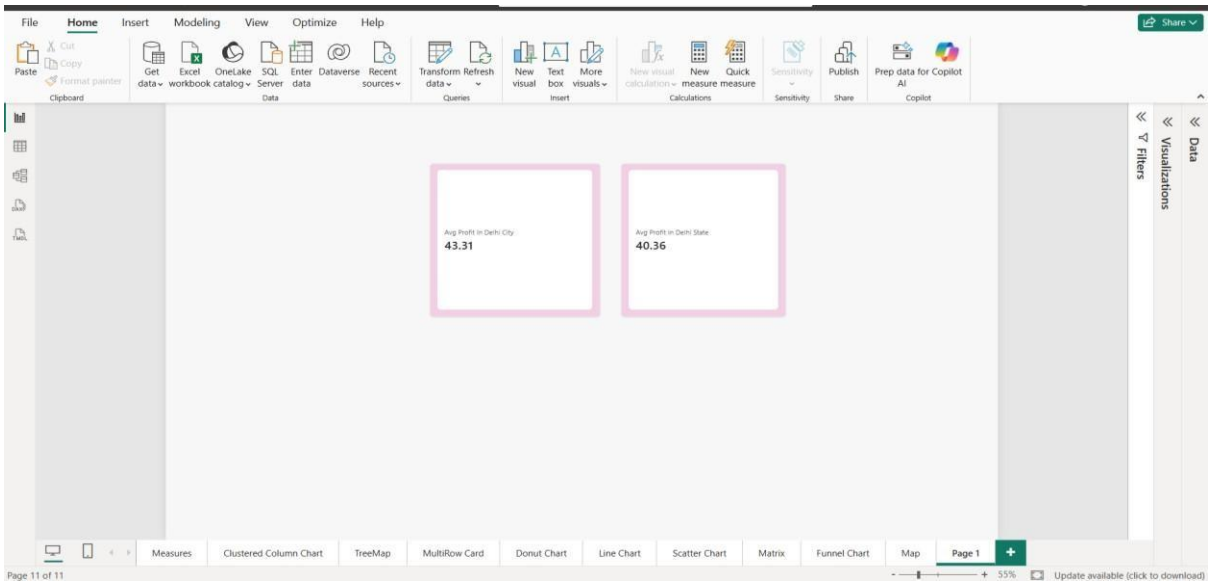


2. CALCULATE AVERAGE PROFIT IN DELHI

In the LIST OF ORDERS Table , DELHI is presented in both CITY and STATE columns. So, 2 measures are calculated.

Avg Profit In Delhi City = CALCULATE(AVERAGE('Order Details'[Profit]) , 'List of Orders'[City] = "Delhi")

Avg Profit In Delhi State = CALCULATE(AVERAGE('Order Details'[Profit]) , 'List of Orders'[State] = "Delhi")



3. CALCULATE YEAR TO DATE (YTD) SALES

Here, the New table named DATE TABLE is created using the NEW TABLE option. This table contains 5 columns named DATE, YEAR, MONTH, MONTH_NAME, MONTH_YEAR.

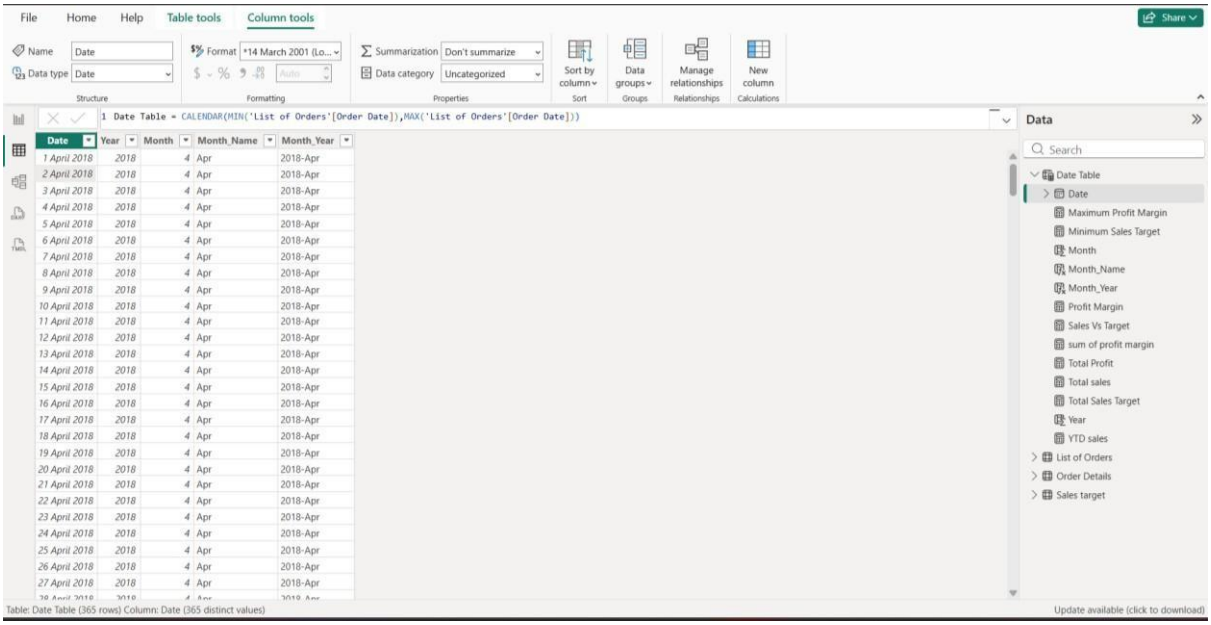
DATE TABLE = CALENDAR(MIN('List of Orders'[Order Date]),MAX('List of Orders'[Order Date]))

MONTH = MONTH('Date Table'[Date])

YEAR = YEAR('Date Table'[Date])

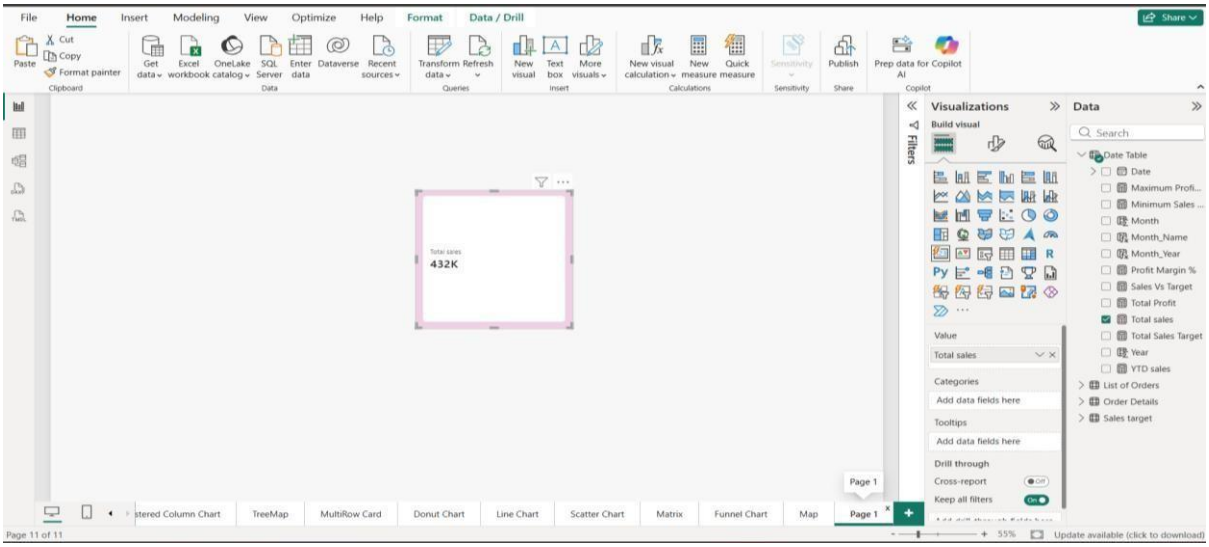
MONTH_NAME = FORMAT('Date Table'[Date], "MMM")

MONTH_YEAR = FORMAT('Date Table'[Date], "YYYY-MMM")



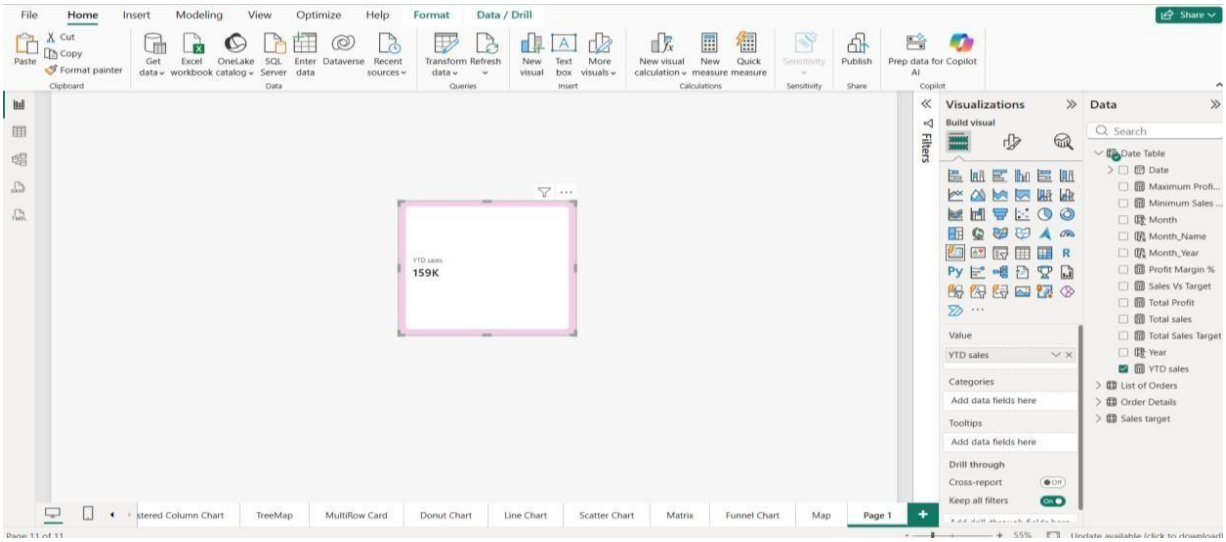
Now, TOTAL SALES is created as a NEW MEASURE.

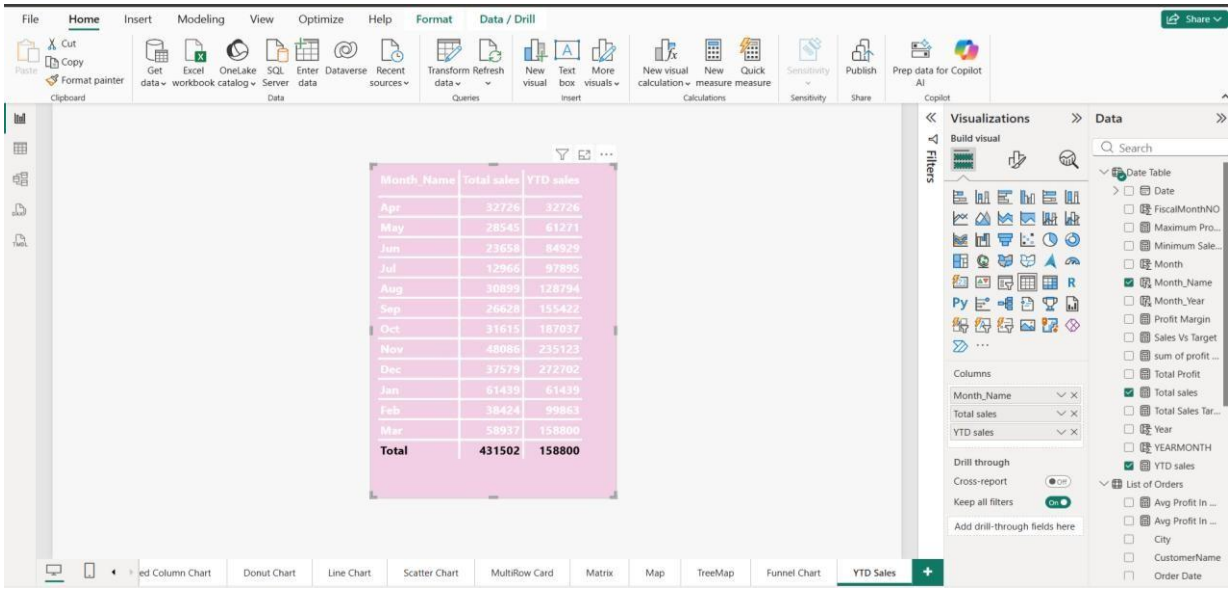
TOTAL SALES = SUM('Order Details'[Amount])



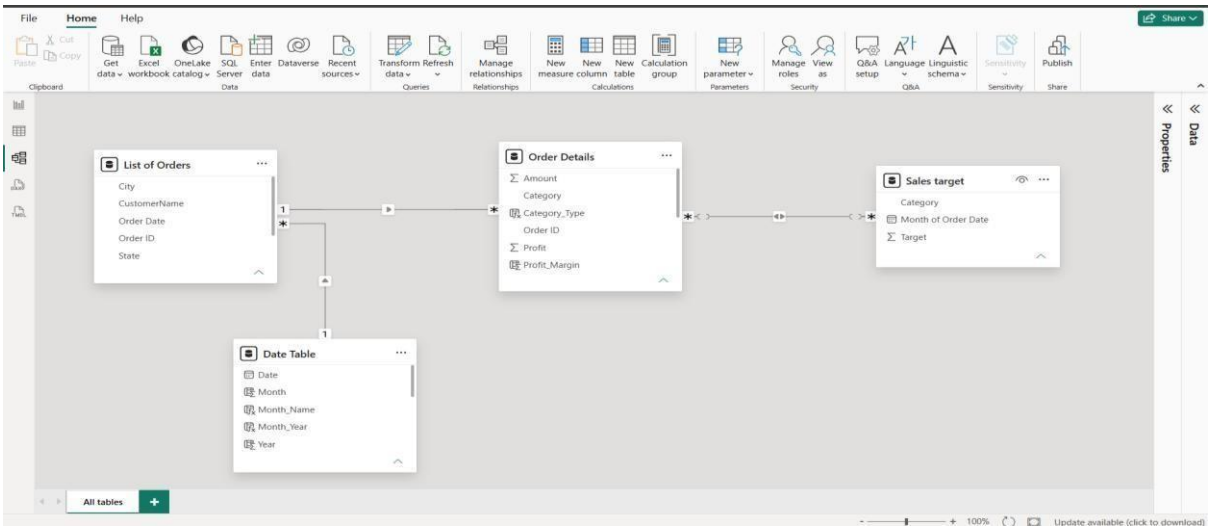
Now , YTD SALES measure is calculated using the formula.

YTD SALES = TOTALYTD([Total sales],'Date Table'[Date])





The Relationship is created between all the 4 tables ., LIST OF ORDERS, ORDER DETAILS, SALES TARGET AND DATE TABLE.



List Of Orders (Order Date) ----- Date Table (Date)

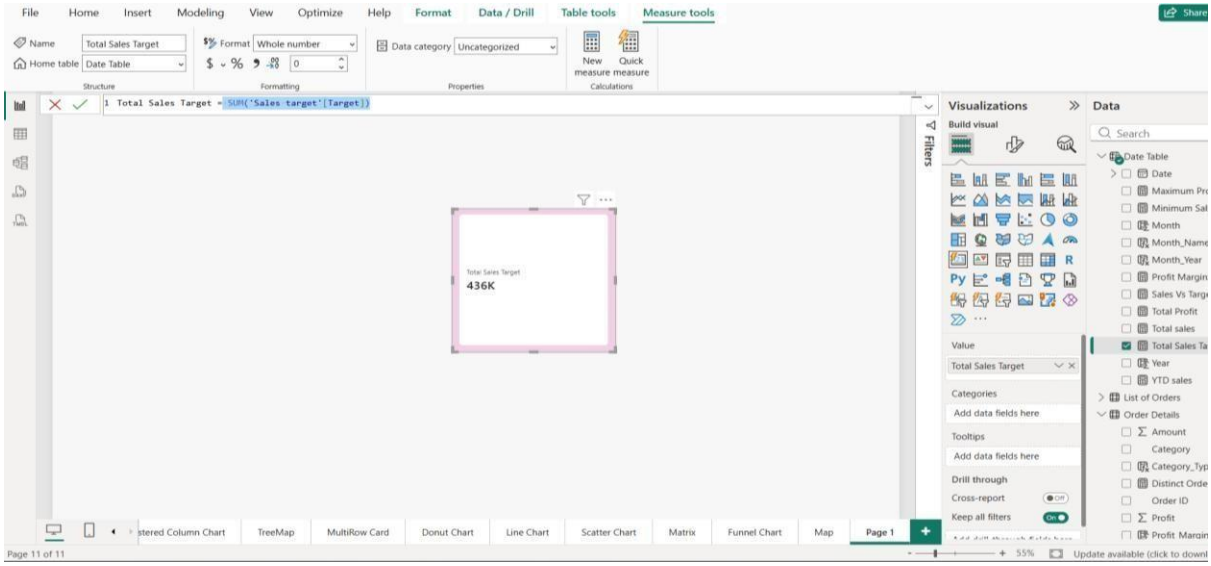
Cardinality = Many to One Cross Filter Directions = Single

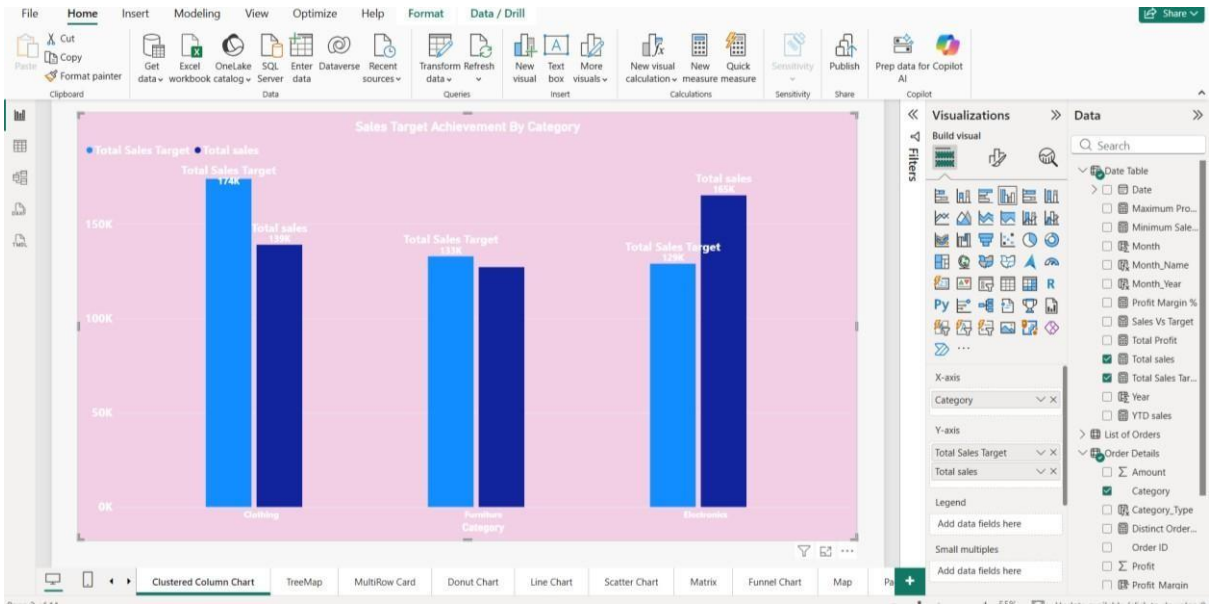
DATA VISUALIZATION

1. **Sales Target Achievement by Category:** Compare actual sales with sales targets by category using a clustered column chart.

Here . new measure TOTAL SALES TARGET is created.

TOTAL SALES TARGET = SUM('Sales target'[Target])





2. **Max Profit Margin by Sub-Category:** Analyze the maximum profit margin for each sub-category of products using a donut chart.

For this chart , the new column PROFIT_MARGIN is created in the ORDER DETAILS table.

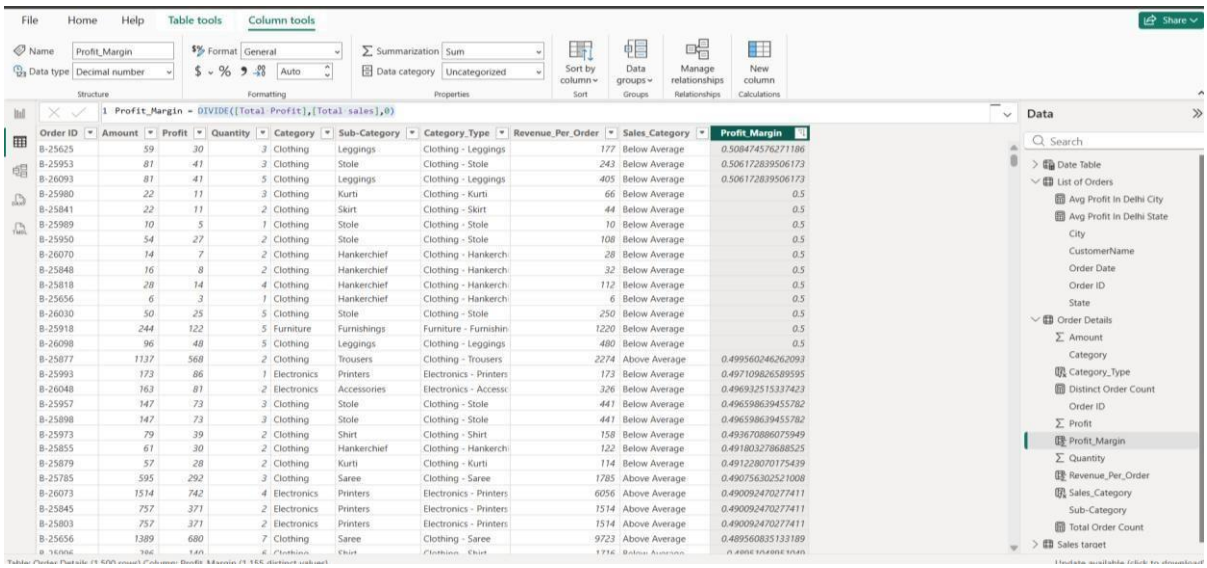
PROFIT_MARGIN = DIVIDE([Total Profit],[Total sales],0)

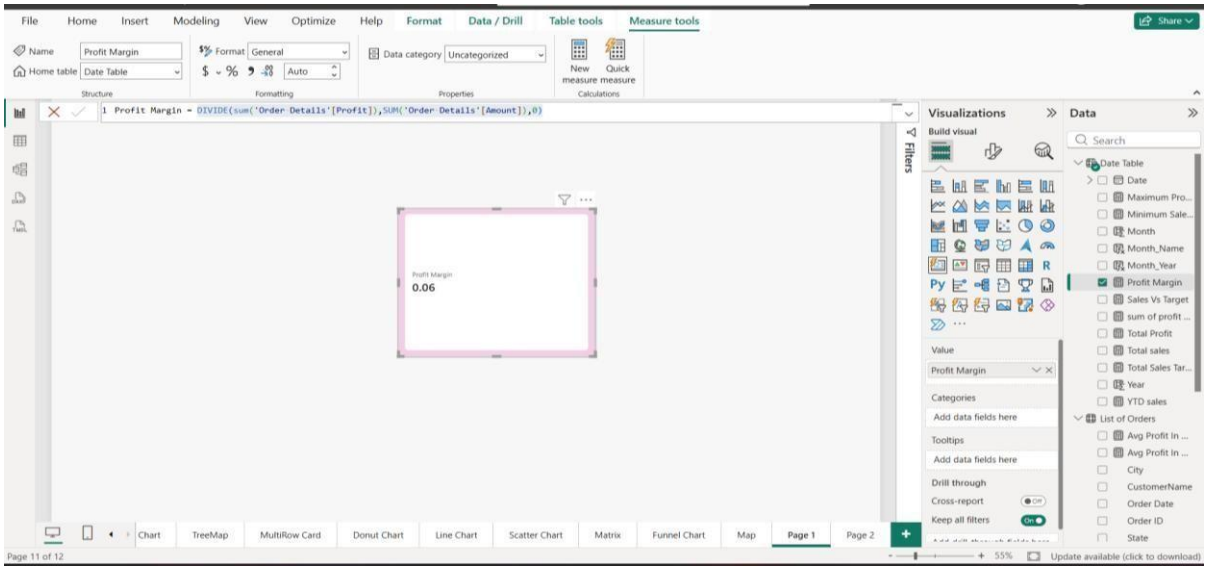
New 3 measures are created.

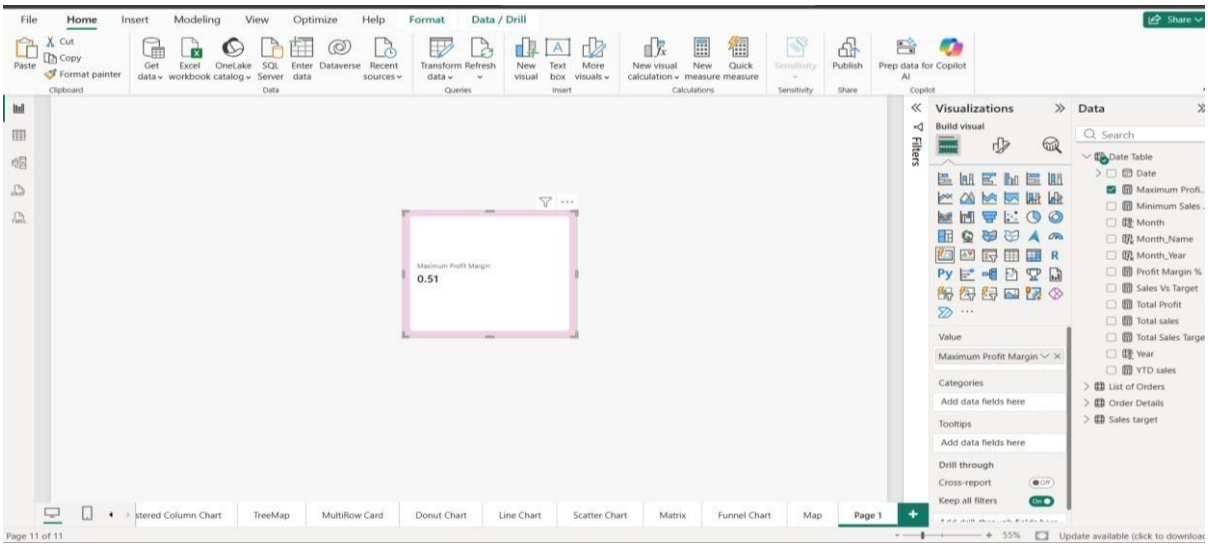
PROFIT MARGIN = DIVIDE (SUM ('Order Details'[Profit]), SUM ('Order Details'[Amount]), 0)

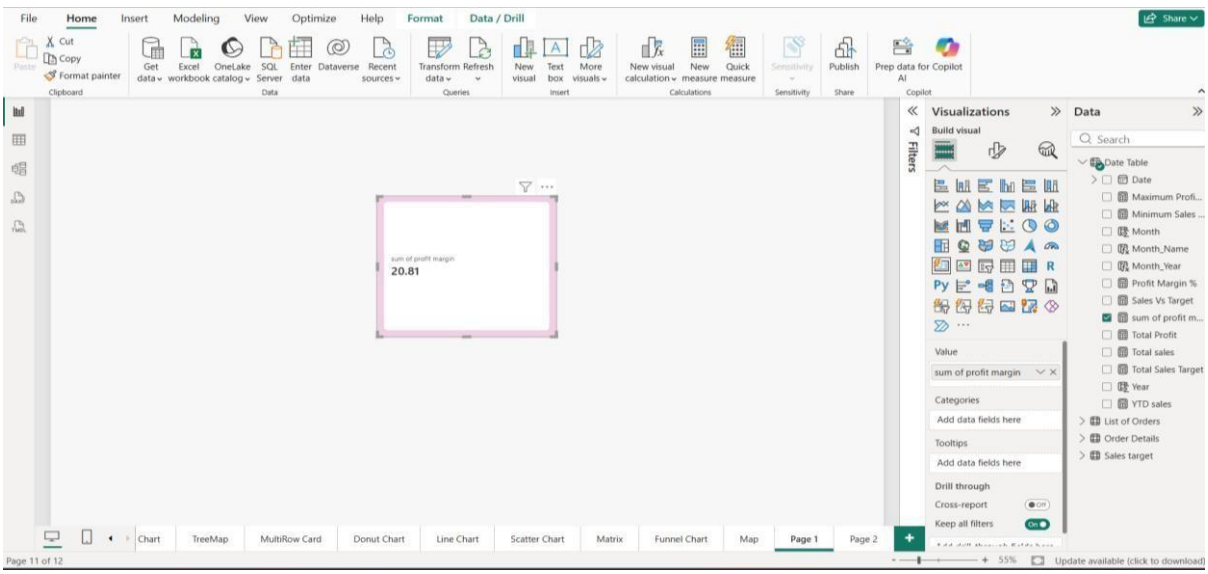
MAXIMUM PROFIT MARGIN = MAX('Order Details'[Profit_Margin]) SUM

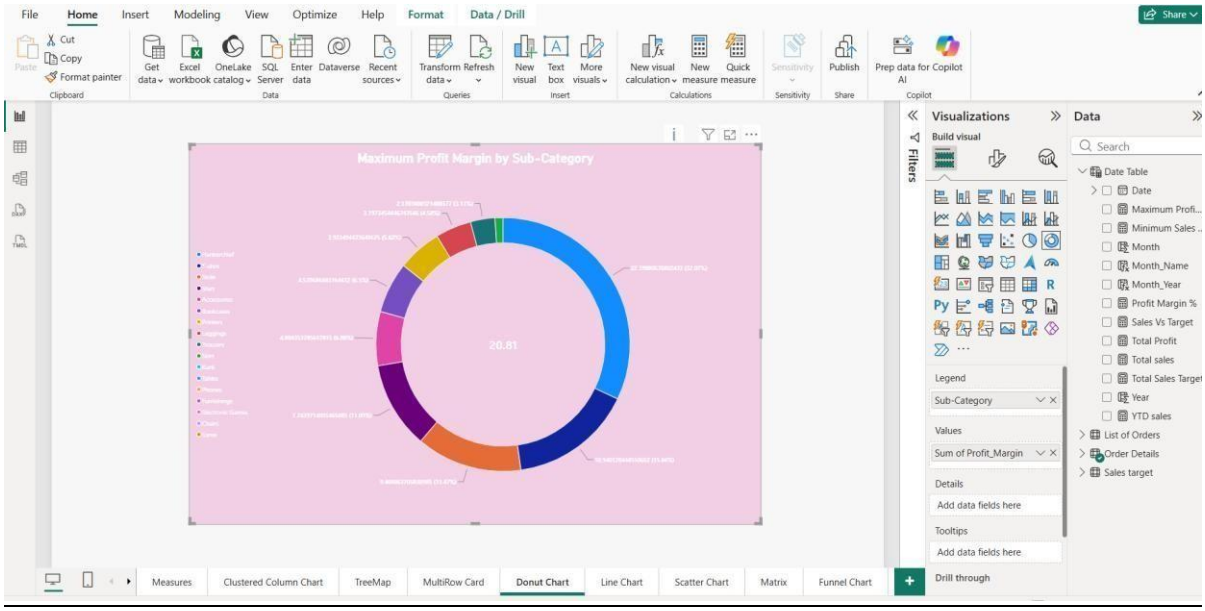
OF PROFIT MARGIN = SUM('Order Details'[Profit_Margin])



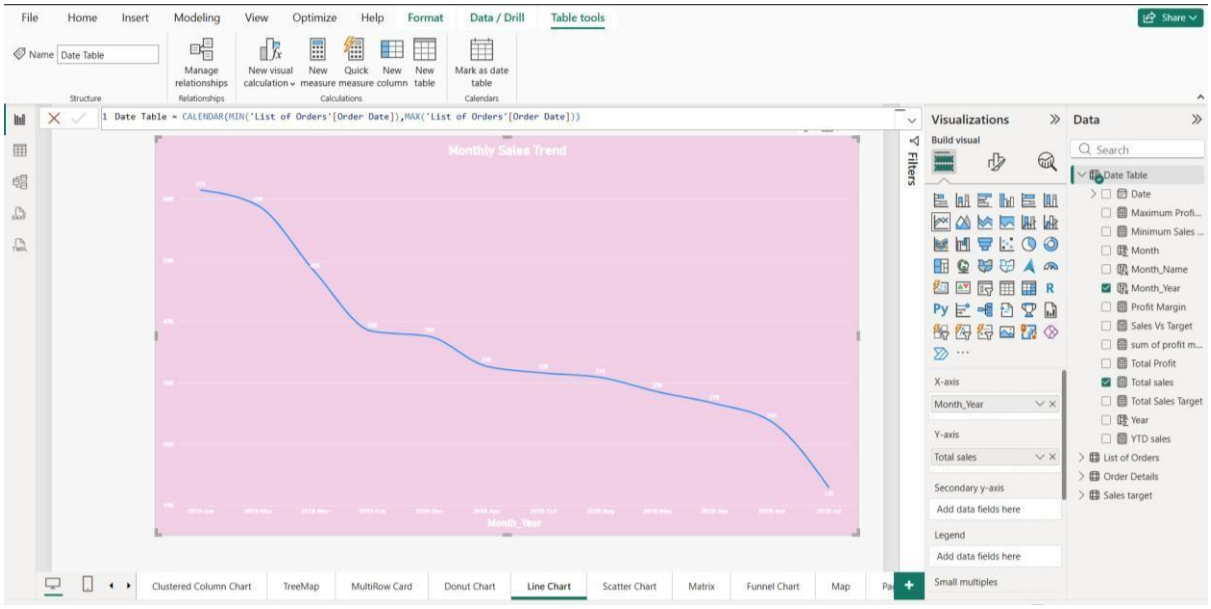

File Home Insert Modeling View Optimize Help Format Data / Drill Table tools Measure tools
Profit Margin = DIVIDE(sum('Order Details'[Profit]),SUM('Order Details'[Amount]),0)
Profit Margin
0.06
Visualizations
Data
Page 11 of 12


File Home Insert Modeling View Optimize Help Format Data / Drill
Maximum Profit Margin
0.51
Visualizations
Data
Page 11 of 11

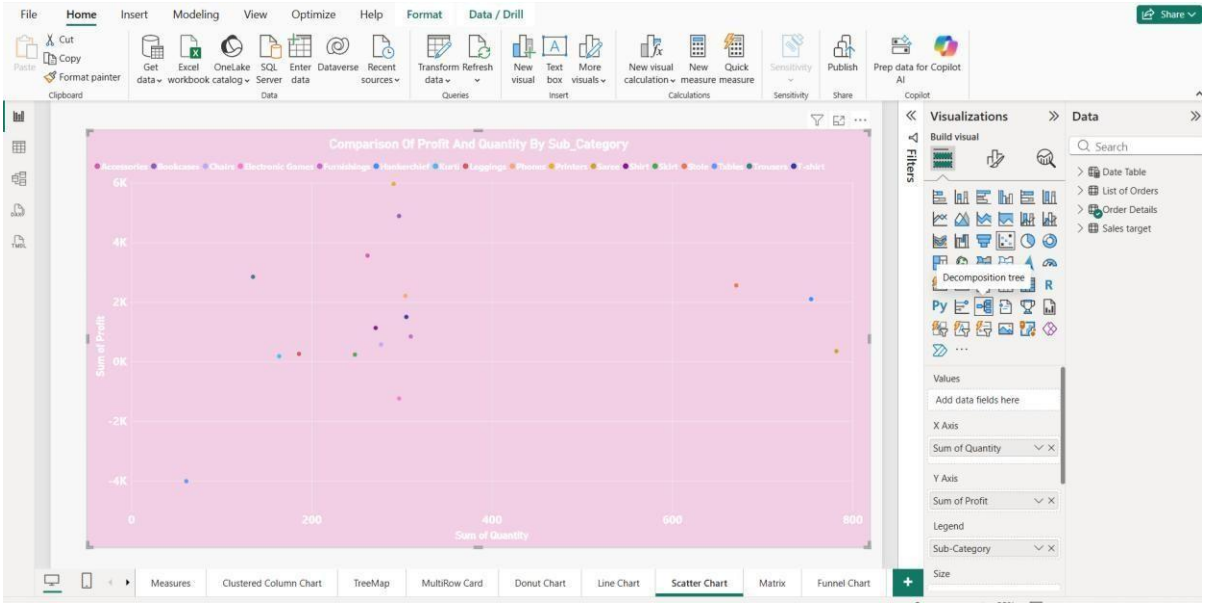

File Home Insert Modeling View Optimize Help Format Data / Drill
sum of profit margin
20.81
Visualizations
Data
Page 11 of 12



3. **Monthly Sales Trend:** Show the trend of monthly sales over time using a line chart

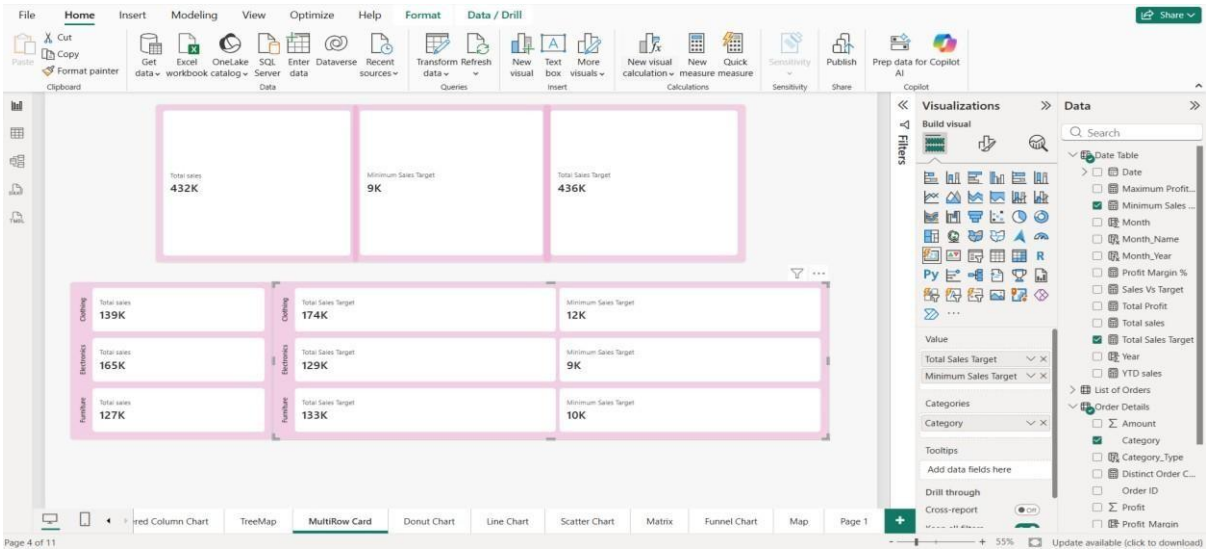


4. **Comparison of Profit and Quantity by Sub-Category:** Compare the relationship between profit and quantity sold for different subcategories using a scatter chart.



5. **Comparison of Total Sales Amount and Target:** Create cards to succinctly display the total sales amount alongside the sales target for quick comparison and analysis. Also, create a multi-row card to display the minimum target for each segment.

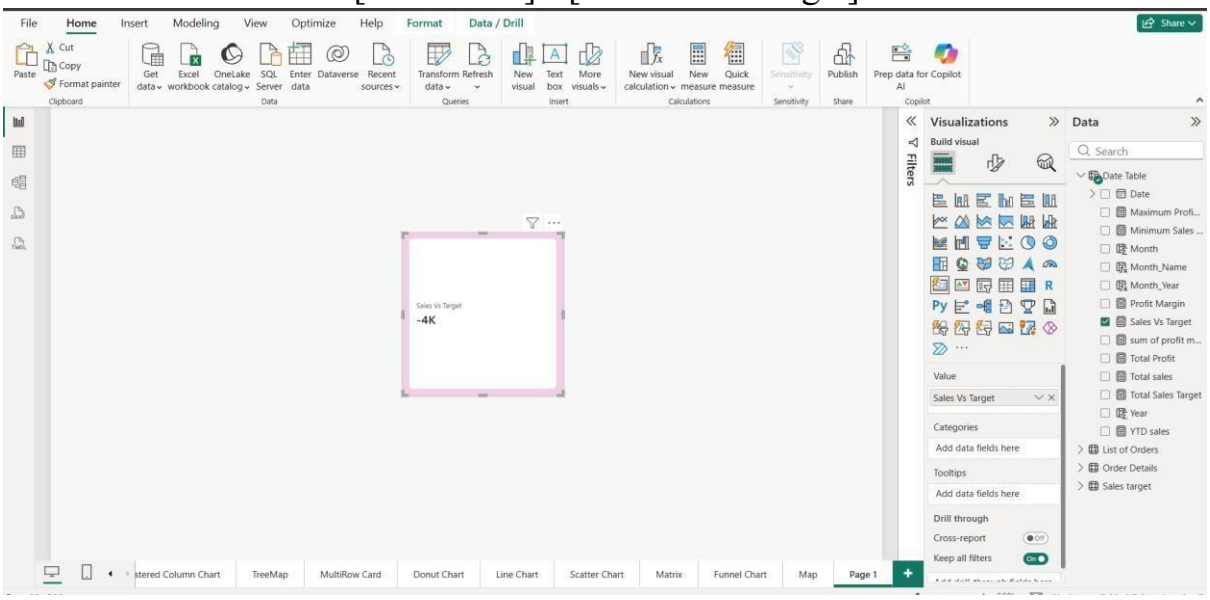
Total Sales Target = SUM('Sales target'[Target])
Minimum Sales Target = MIN('Sales target'[Target])

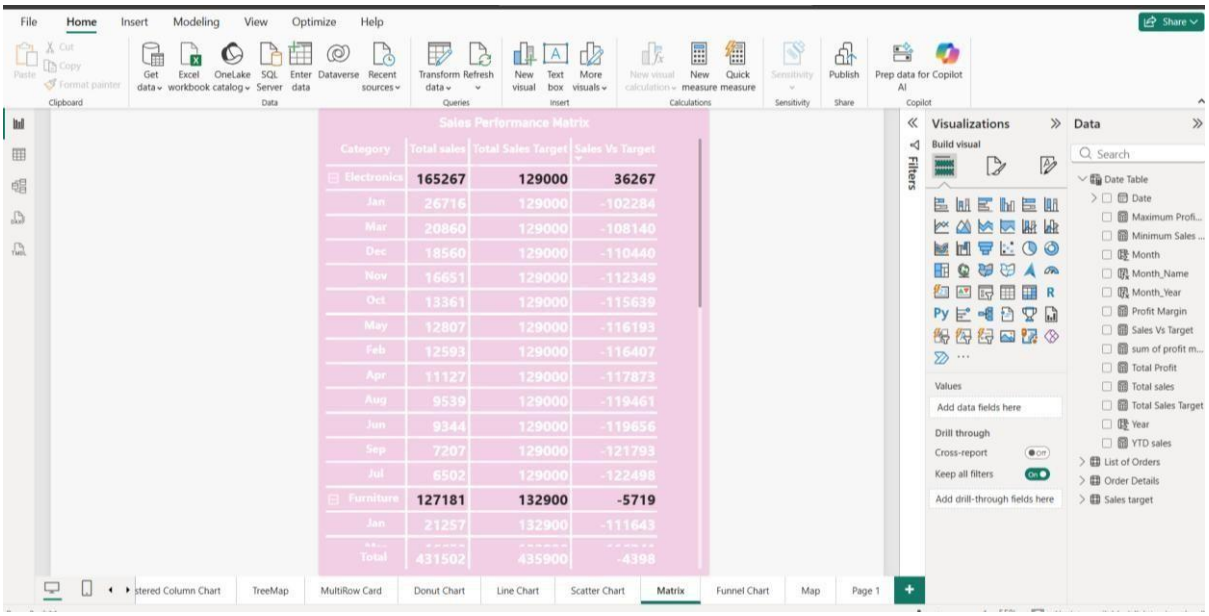


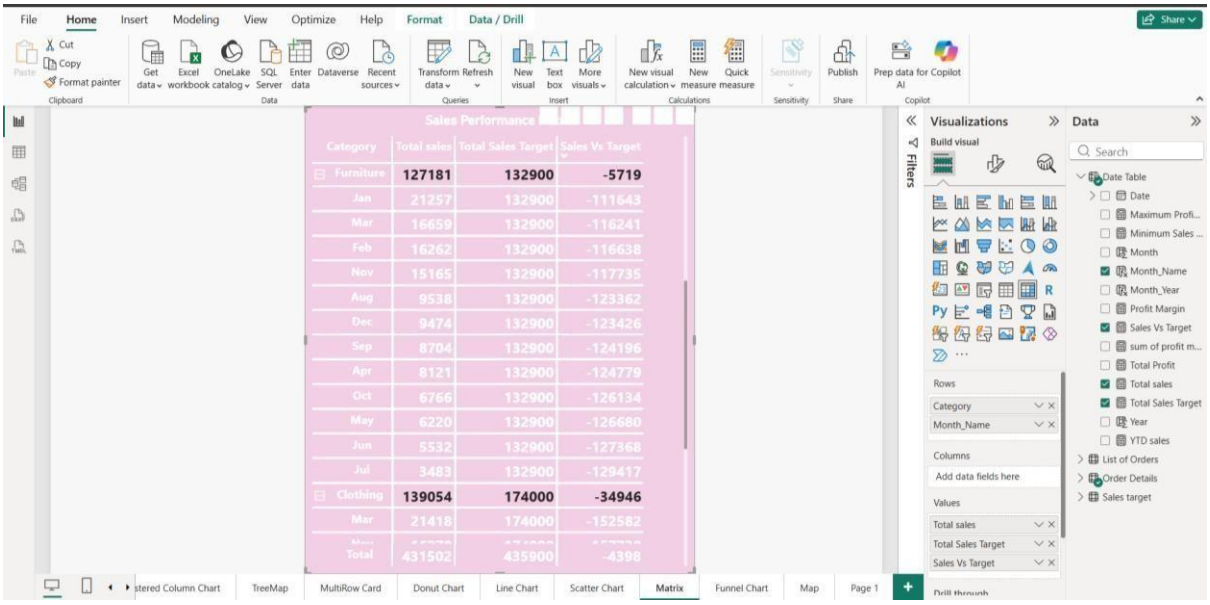
6. **Sales Performance Matrix:** Build a matrix view to analyze how actual sales compare to sales targets across different categories and months.

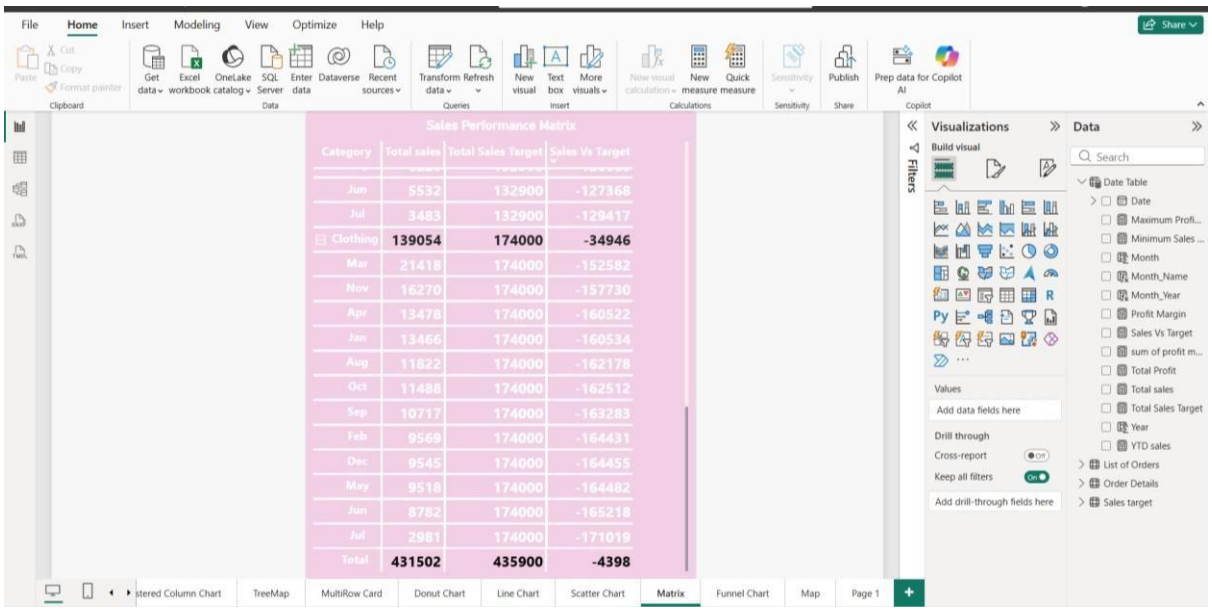
The new measure is created.

SALES VS TARGET = [Total sales] - [Total Sales Target]

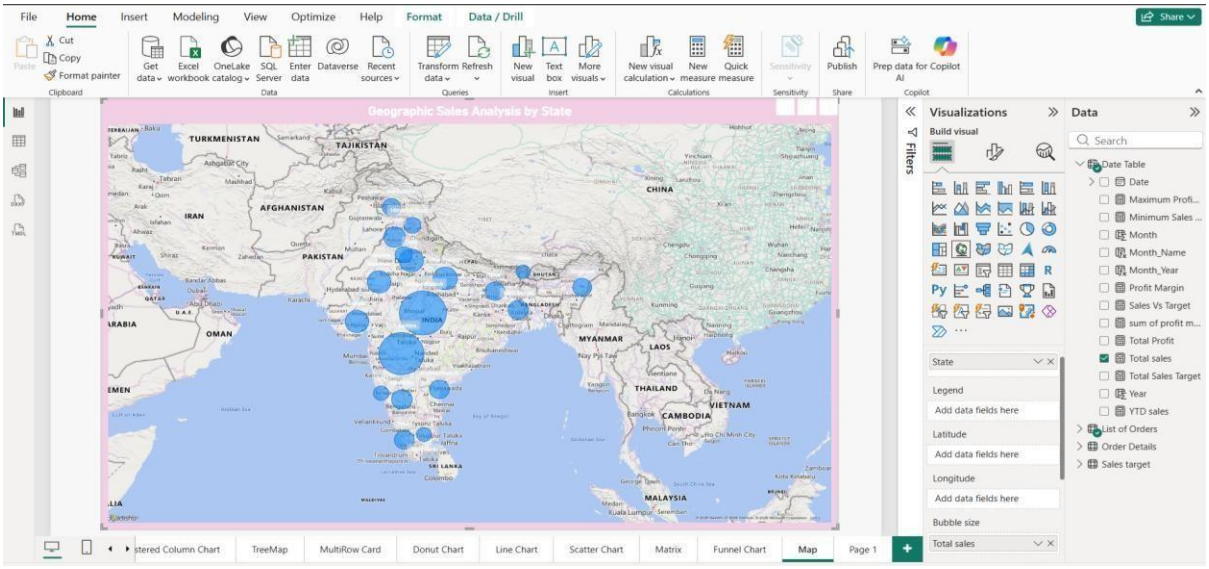




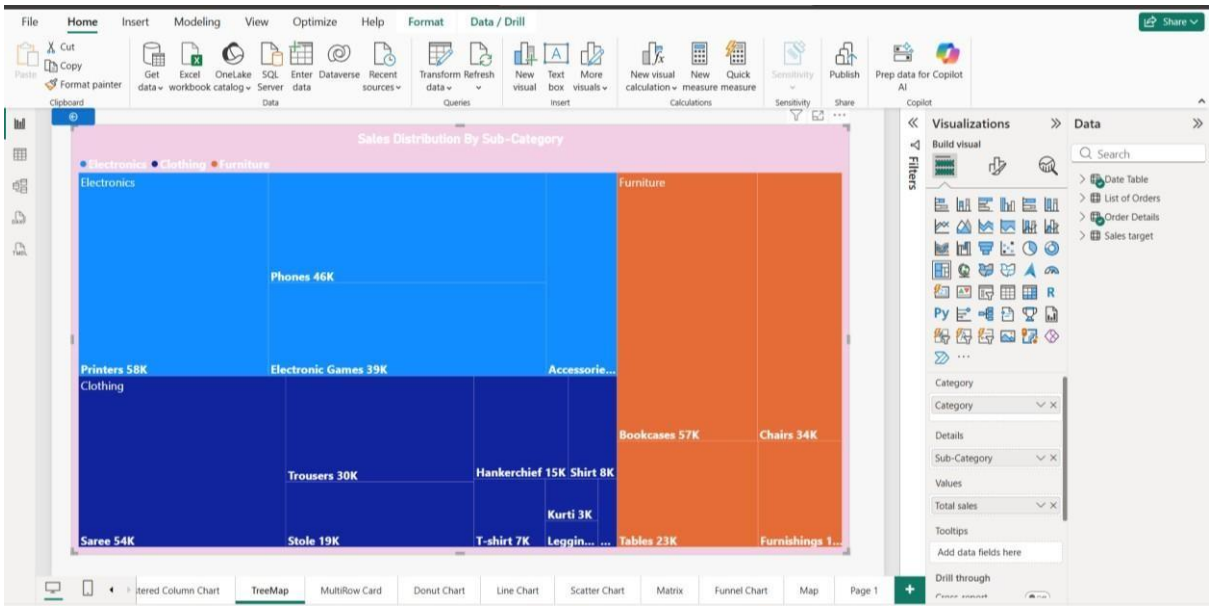




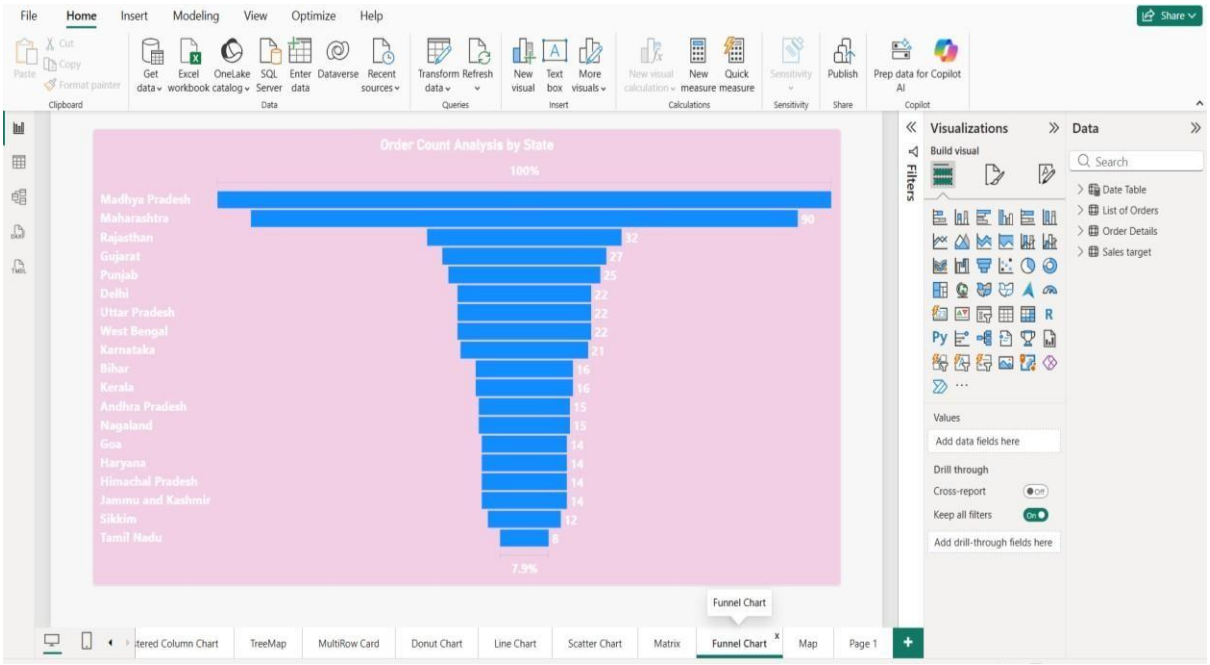
7. **Geographic Sales Analysis:** Visualize total sales on a map by city to identify regional sales patterns.



8. **Sales Distribution by Sub-Category:** Represent the sales distribution across different sub-categories using a Treemap.



9. **Order Count Analysis by State:** Create a funnel chart to visualize the distribution of order counts across different states.


File
Home
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Modeling
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Optimize
Help
Share
Order Count Analysis by State
100%
Madhya Pradesh
Maharashtra
Rajasthan
Gujarat
Punjab
Delhi
Uttar Pradesh
West Bengal
Karnataka
Bihar
Kerala
Andhra Pradesh
Nagaland
Goa
Haryana
Himachal Pradesh
Jammu and Kashmir
Sikkim
Tamil Nadu
90
32
27
25
22
22
22
21
16
16
15
15
14
14
14
14
12
8
7.9%
Funnel Chart
Visualizations
Build visual
Values
Add data fields here
Drill through
Cross-report
Keep all filters
Add drill-through fields here
Data
Search
Date Table
List of Orders
Order Details
Sales target
Filters
tered Column Chart
TreeMap
MultiRow Card
Donut Chart
Line Chart
Scatter Chart
Matrix
Funnel Chart
Map
Page 1